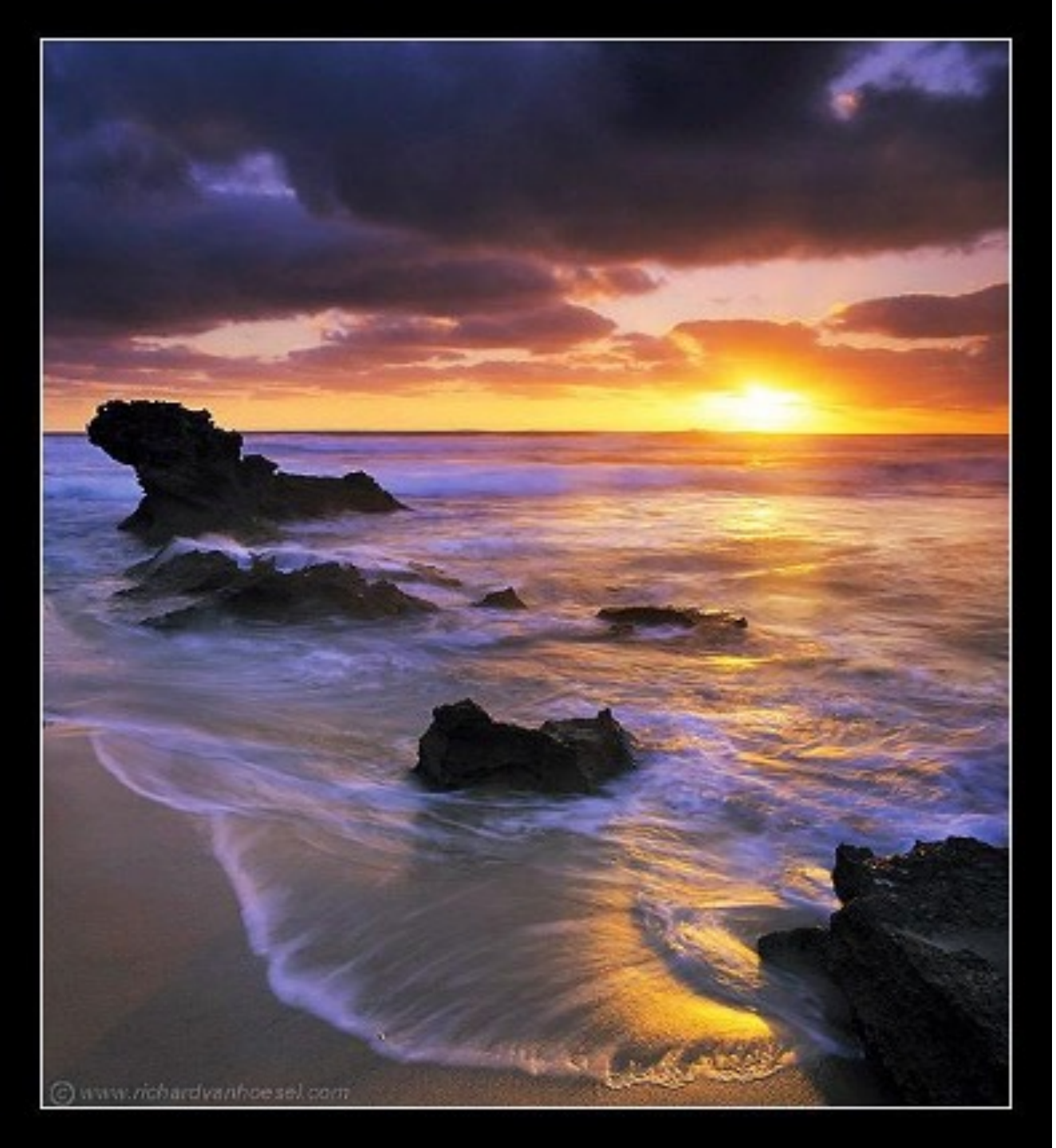


上海科技大学
ShanghaiTech University

Light and Shading

CS172 Computer Vision I

Instructor: Jiayuan Gu



What determines a pixel's intensity?

What can we infer about the scene from pixel intensities?

Image Source: A. Efros

Agenda

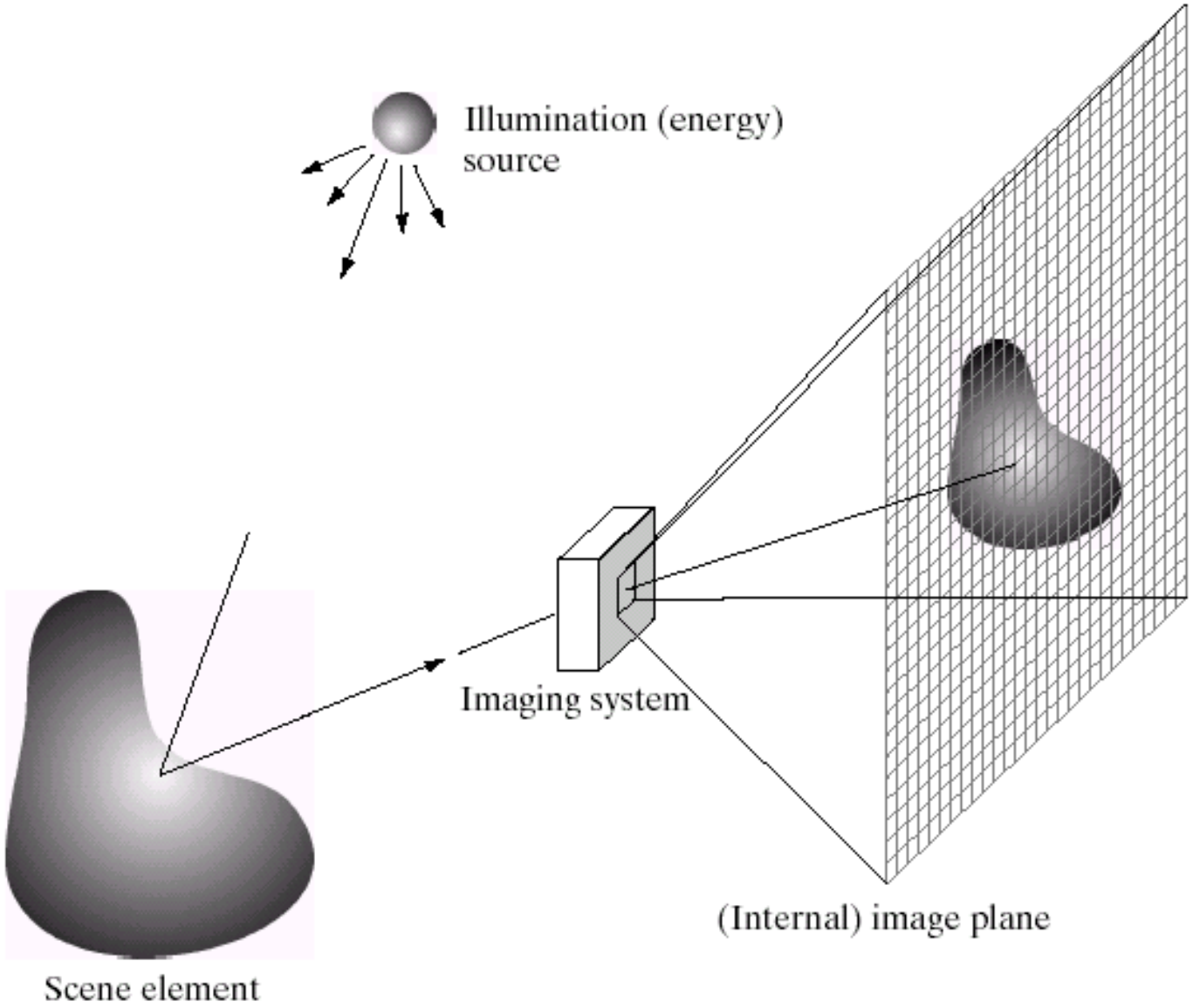
- Roadmap
  - From light to pixels
    - Radiance and irradiance
    - Reflectance and shading
    - Digital camera
  - From pixels to shape: photometric stereo

- Objectives
  - explain what determines image brightness;
  - derive the Lambertian photometric-stereo system;
  - identify when the model will fail.

Photometric Image Formation

Part I: From Scene Light to Pixel Values

Image Formation



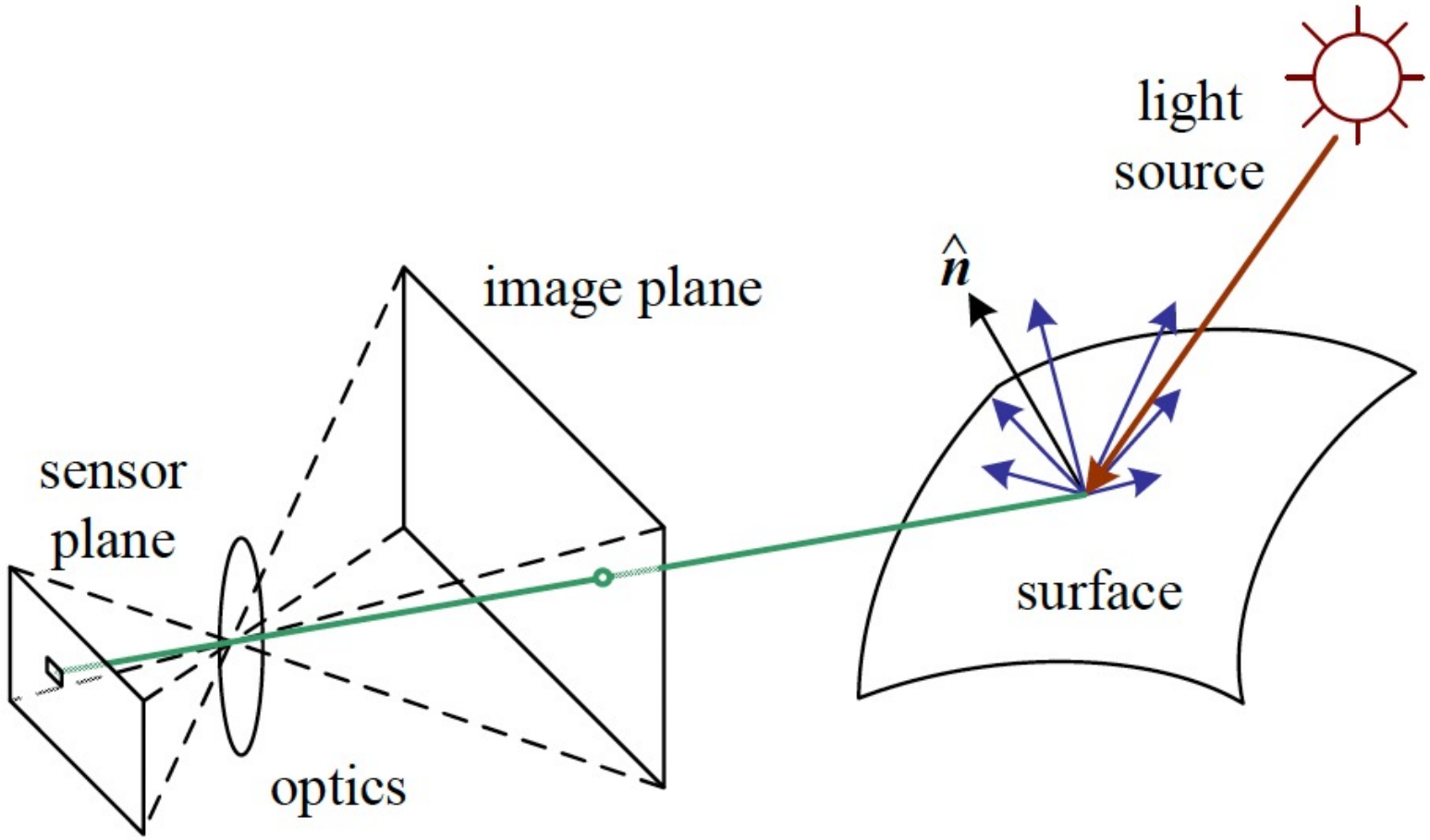
Last lecture: geometry determines where the scene point projects
This lecture: radiometry determines the measured intensity

What Determines Brightness?

Scene
illumination
surface orientation
material/reflectance

Camera optics
aperture
exposure

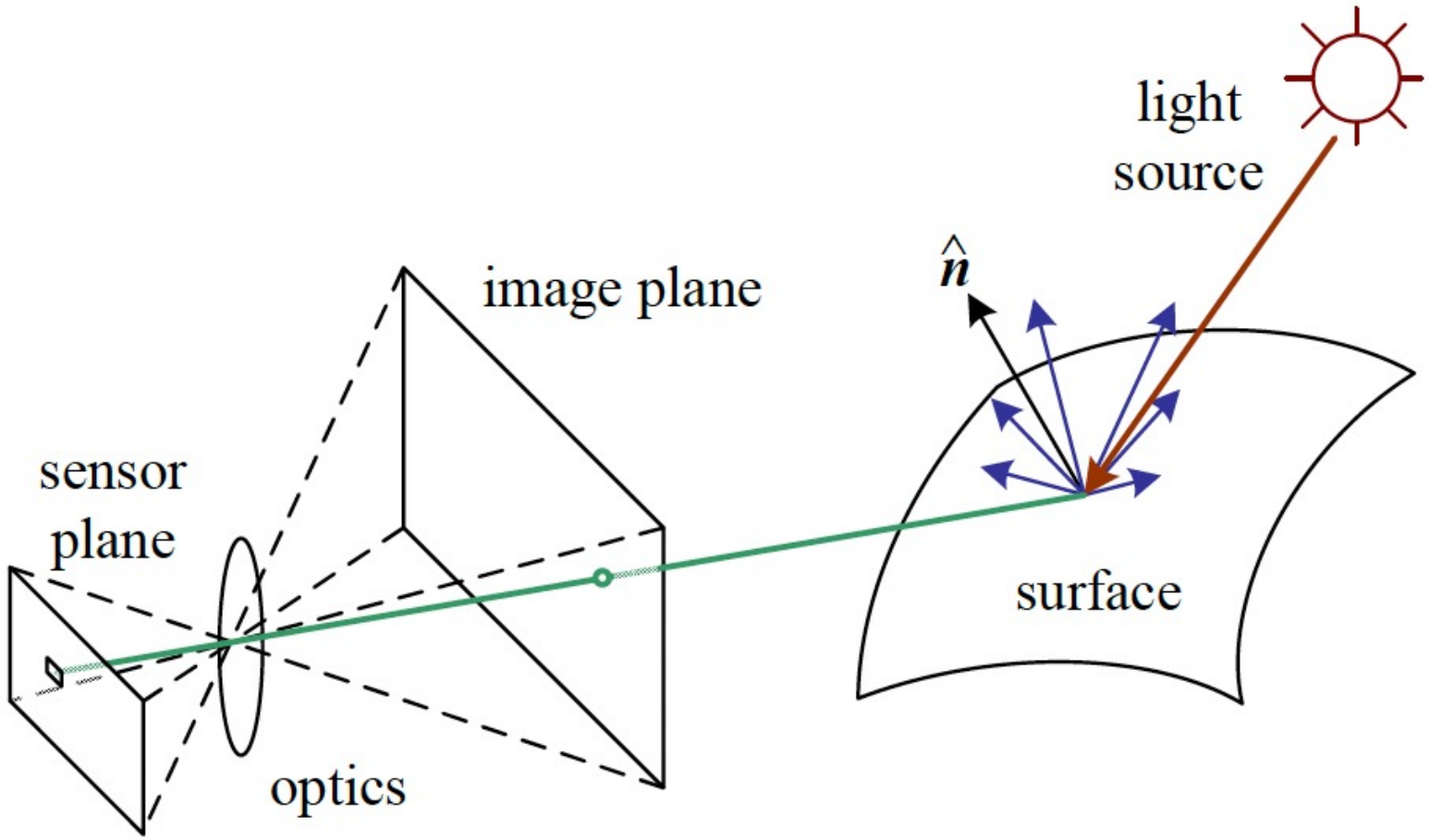
Sensor/processing
sensor response



The intensity value is not an intrinsic property of the object

Szeliski, Figure 2.14

Light is Reflected and Reaches the Sensor



Let's first check the green path between the surface and the sensor plane

Szeliski, Figure 2.14

The Intensity Depends on Received Light Energy

L: *Radiance* emitted from ***P*** toward ***P'***
Directional light traveling through space
Unit: Watts per square meter per *steradian*

E: *Irradiance* falling on ***P'*** from the lens
Energy arriving at a surface
Unit: Watts per square meter

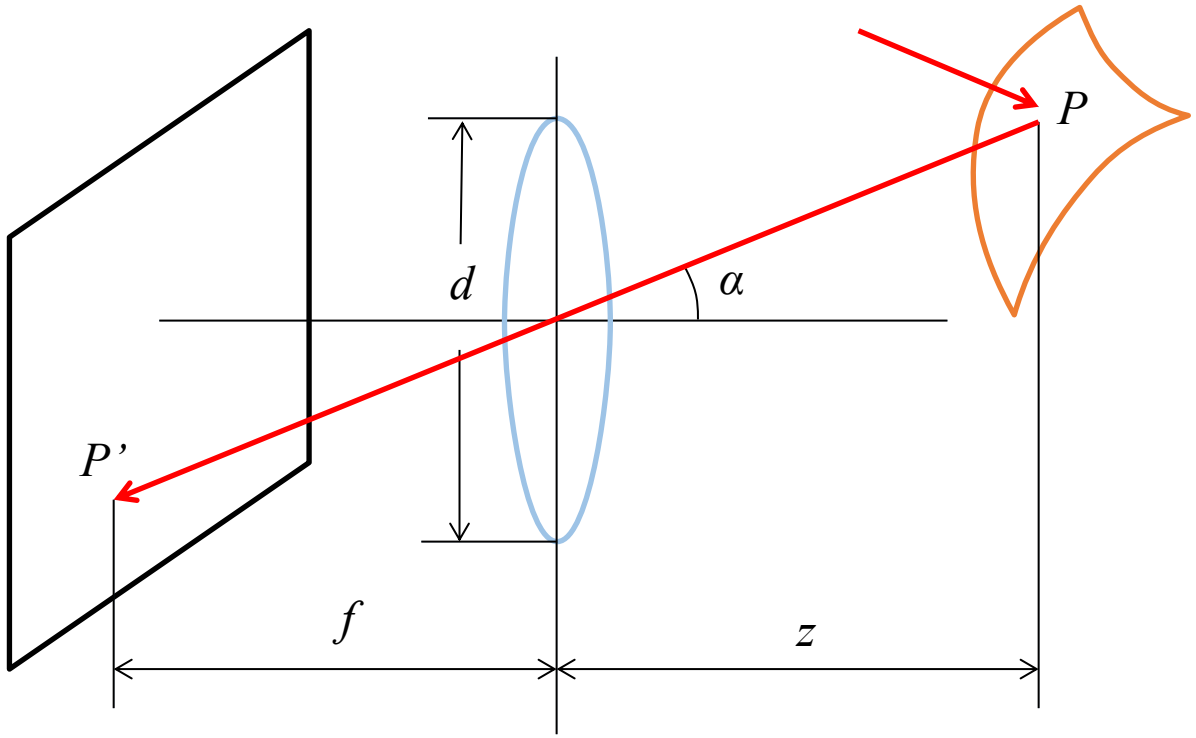
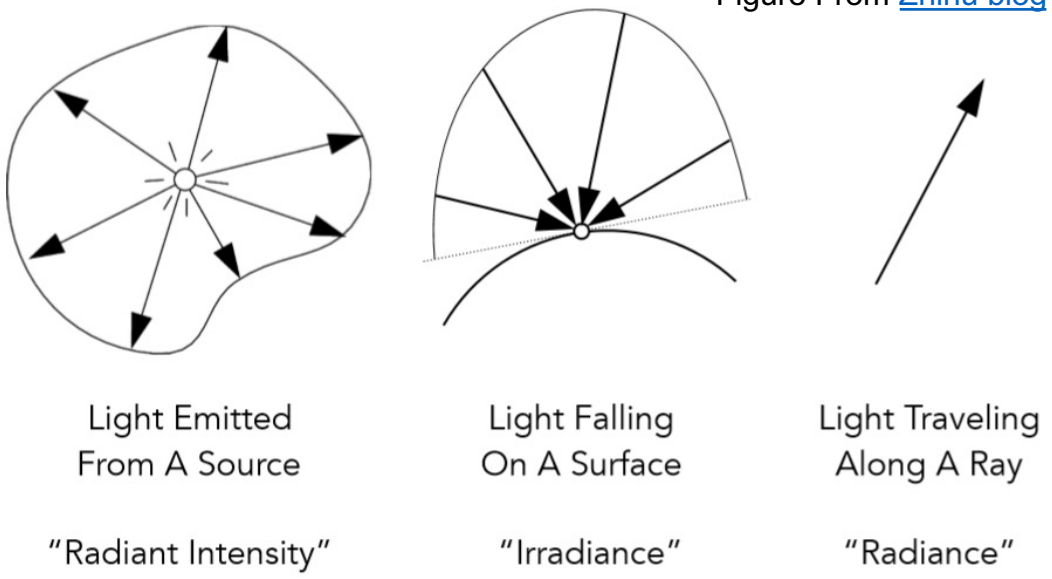


Figure From Zhihu blog

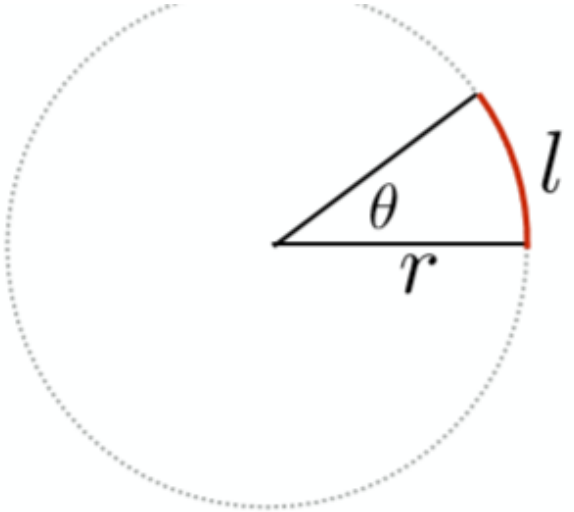


Szeliski, Ch 2.2.3

Steradian: Unit of Solid Angle

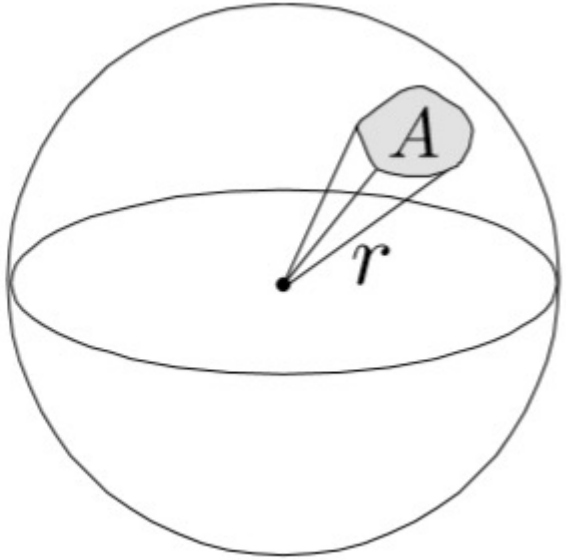
Angle: ratio of subtended arc length on circle to radius

- $\theta = \frac{l}{r}$
- **Circle has 2π radians**



Solid angle: ratio of subtended area on sphere to radius squared

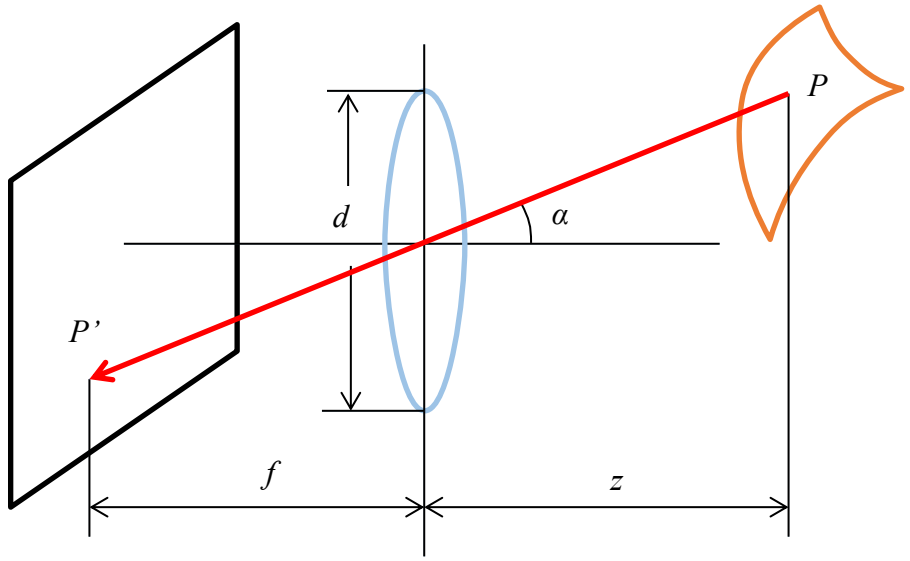
- $\Omega = \frac{A}{r^2}$
- **Sphere has 4π steradians**



Self-reading

From Zhihu blog

Fundamental Radiometric Relation



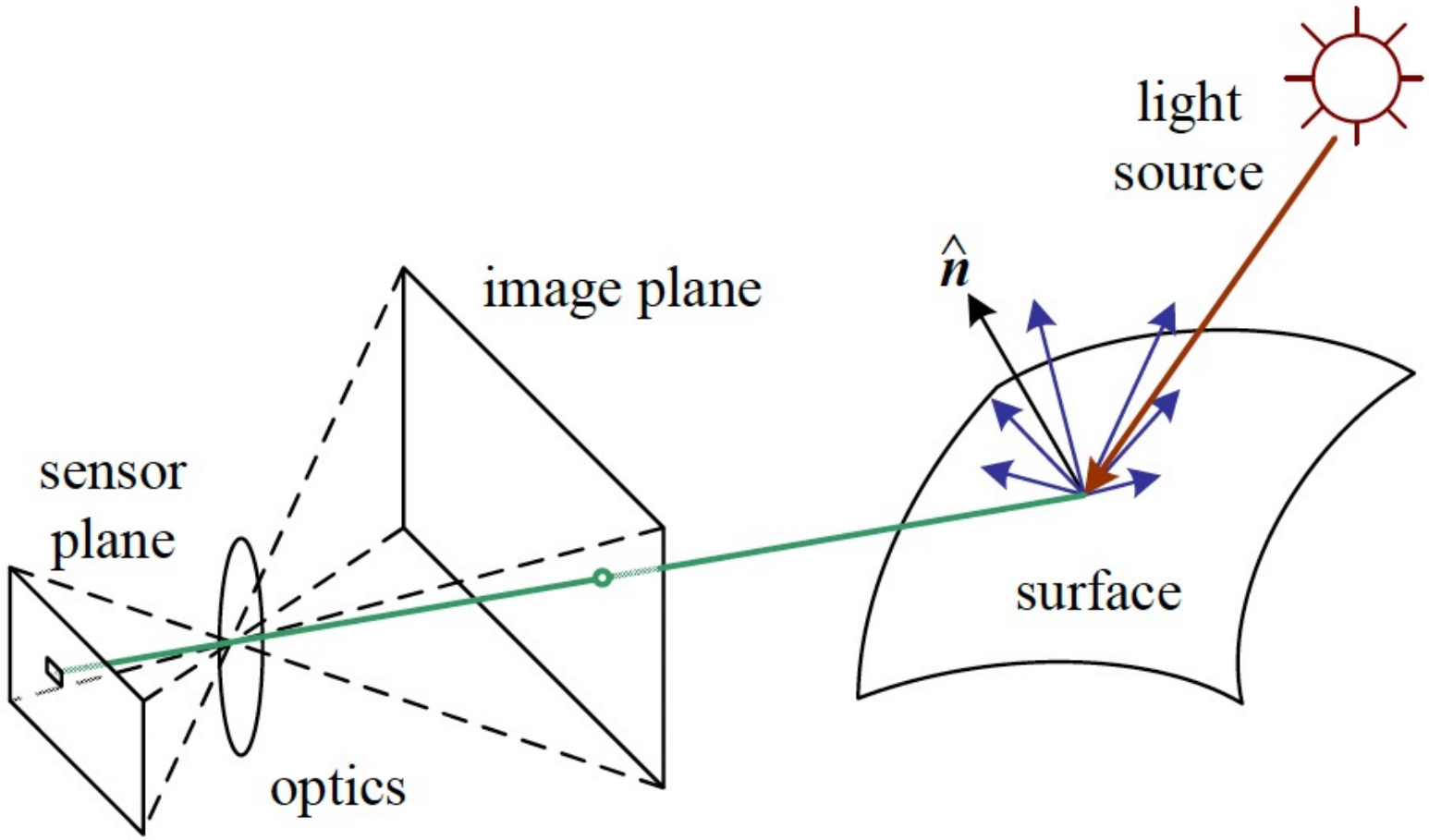
The relation between radiance and irradiance

$$E = \left[\frac{\pi}{4}\left(\frac{d}{f}\right)^2 \cos^4 \alpha\right] L$$

- Image irradiance is linearly related to scene radiance L
- Irradiance is proportional to **the area of the lens** and inversely proportional to **the squared distance between the lens and the image plane**
- Irradiance falls off as **the angle α between the viewing ray and the optical axis** increases (*natural vignetting*)

Szeliski, Ch 2.2.3

Light Reaches the Surface and then is Reflected



Let's next check the incident light and reflected light

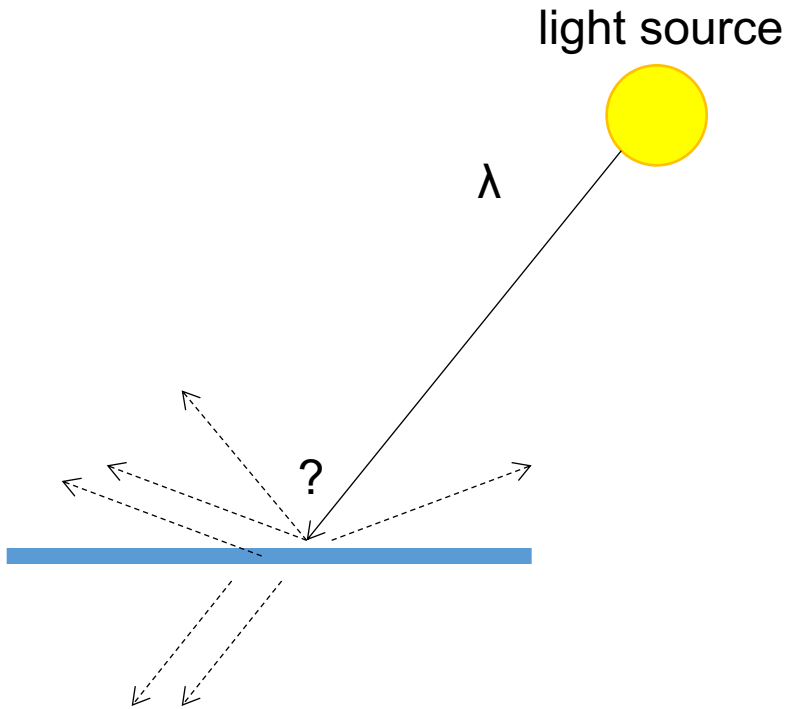
Szeliski, Figure 2.14

Reflectance and Shading

Given incoming light, how much leaves toward the camera?

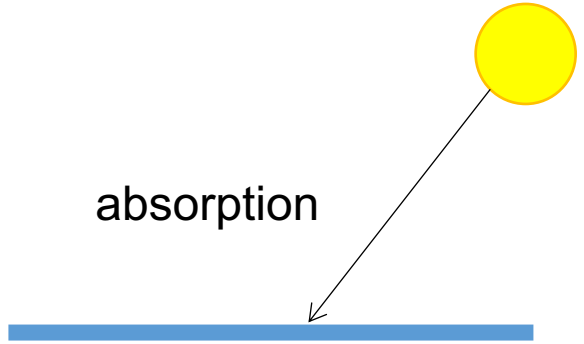
A Photon's Life Choices

- Absorption
- Reflection
  - Diffuse
  - Specular
- Transmission
- Other / nonlocal effects
  - Fluorescence
  - Phosphorescence
  - Subsurface scattering
  - Interreflection

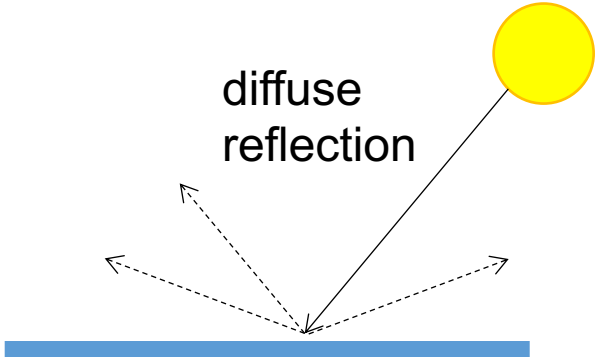


Reflection

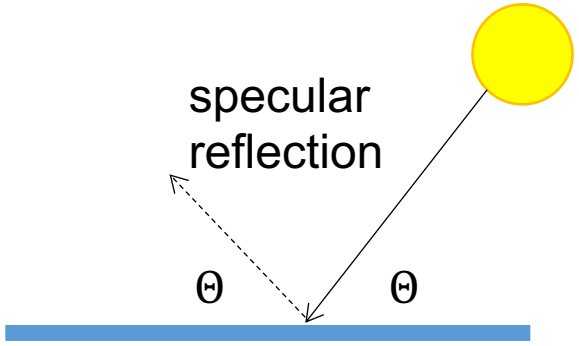
- For an opaque surface, some incident energy is absorbed and some is reflected



- **(Lambertian) diffuse reflection:** outgoing radiance is independent of viewing direction



- **Specular reflection**: light bounces off (like a mirror), depending on viewing direction





Youtube: How to make Blender Glossy Plastic and Matte Plastic Material using BSDF shader

Shading

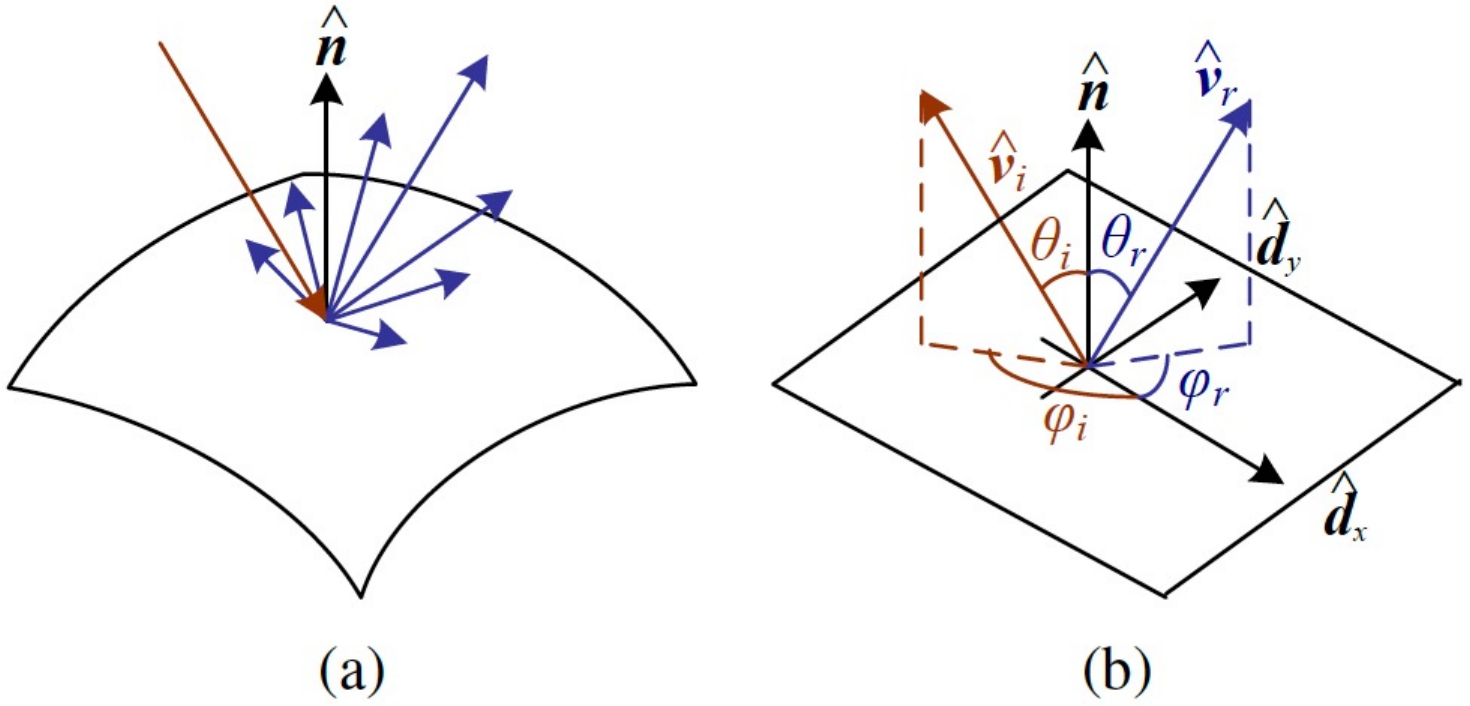
- In computer graphics, *shading* refers to the process of **altering the color** of an object in the 3D scene, based on things like, but not limited to, the **surface's angle to lights**, **its distance from lights**, **its angle to the camera** and **material properties** to create a photorealistic effect

- Not to be confused with techniques of adding shadows



Photo courtesy of the Caltech Vision Lab

Bidirectional Reflectance Distribution Function (BRDF)



(a) (b)

$\hat{v}_i$: incident direction
$\hat{v}_r$: reflected direction
$\hat{n}$: normal direction
λ: wavelength

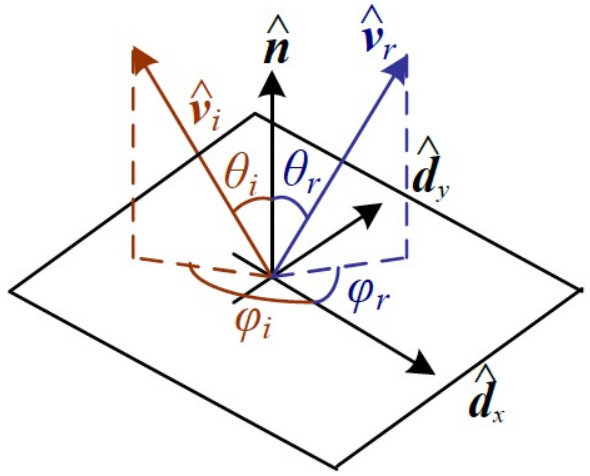
$$f_r(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}}; \lambda)$$

(isotropic surface)

How a material converts incoming light from one direction into outgoing light toward another direction

Szeliski, Ch 2.2.2, Figure 2.15

Bidirectional Reflectance Distribution Function (BRDF)



The amount of light exiting a surface point in a direction $\hat{v}_r$

$$L_r(\hat{\mathbf{v}}_r;\lambda) = \int L_i(\hat{\mathbf{v}}_i;\lambda) f_r(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}};\lambda) \cos^+\theta_i \, d\hat{\mathbf{v}}_i$$

incoming light

foreshortening factor

$$\cos^+\theta_i = \max(0, \cos\theta_i)$$

$$f_r(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}};\lambda)$$

Pixel brightness entangles illumination, material, and geometry

$\hat{v}_i$: incident direction
$\hat{v}_r$: reflected direction
$\hat{n}$: normal direction
λ: wavelength

For a finite number of (point) light sources

$$L_r(\hat{\mathbf{v}}_r;\lambda) = \sum_i L_i(\lambda) f_r(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}};\lambda) \cos^+\theta_i$$

Szeliski, Ch 2.2.2

Lambertian Diffuse Reflection

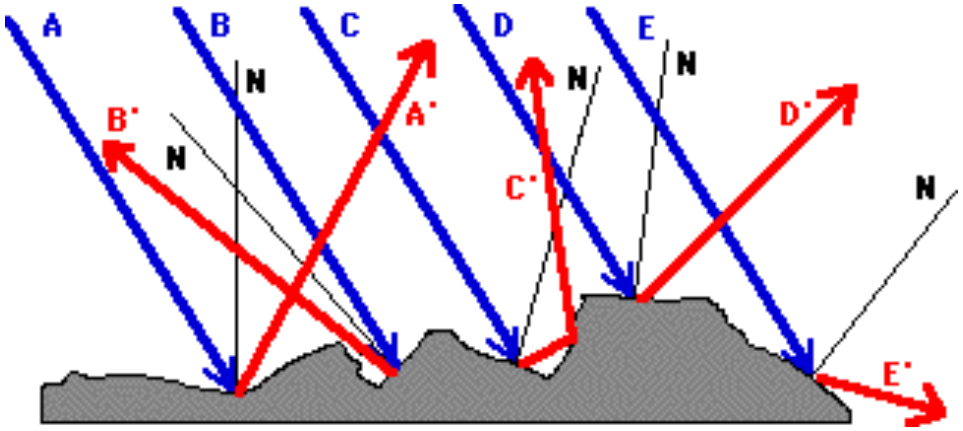
$$f_d(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}}; \lambda) = f_d(\lambda)$$

The BRDF is constant

$$L_d(\hat{\mathbf{v}}_r; \lambda) = \sum_i L_i(\lambda) f_d(\lambda) \cos^+ \theta_i = \sum_i L_i(\lambda) f_d(\lambda) [\hat{\mathbf{v}}_i \cdot \hat{\mathbf{n}}]^+$$

Shading equation

Why does a rough surface diffuse a beam of light?

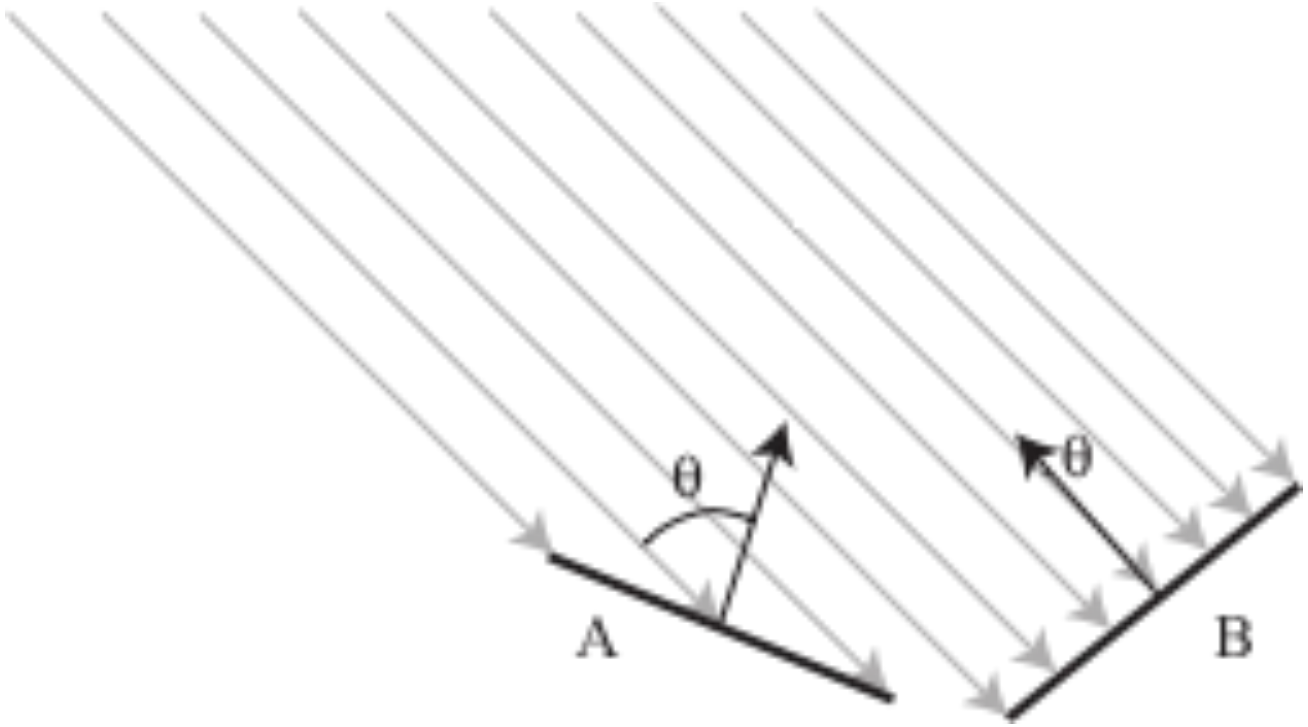


https://www.physicsclassroom.com/class/refln/lesson-1/specular-vs-diffuse-reflection

Lambert's Cosine Law

$$I(x) = \rho(x)(\boldsymbol{S} \cdot \boldsymbol{N}(x))$$

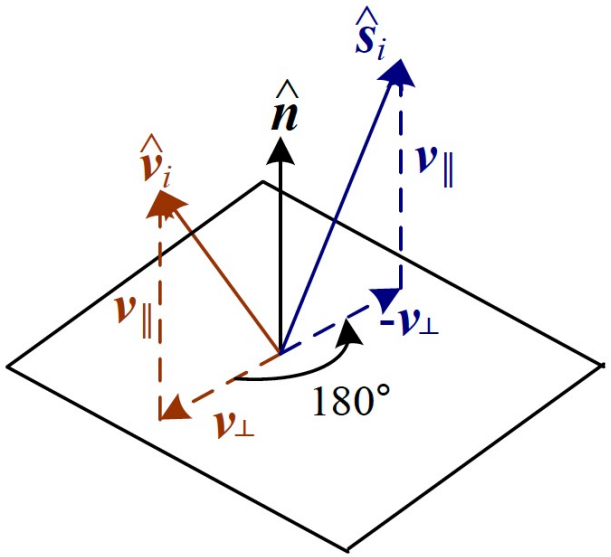
ρ: albedo
$\boldsymbol{S}$: light source direction
$\boldsymbol{N}$: surface normal
I: reflected intensity



The relation between intensity and surface orientation

Credit to David Forsyth

Specular Reflection



specular reflection direction

$$\hat{\mathbf{s}}_i = \mathbf{v}_\parallel - \mathbf{v}_\perp = (2\hat{\mathbf{n}}\hat{\mathbf{n}}^T - \mathbf{I})\mathbf{v}_i.$$

The Phong model uses a power of the cosine of the angle

$$f_s(\theta_s;\lambda) = k_s(\lambda)\cos^{k_e}\theta_s$$

The Torrance and Sparrow micro-facet model uses a Gaussian

$$f_s(\theta_s;\lambda) = k_s(\lambda)\exp(-c_s^2\theta_s^2)$$

Direct vs. Indirect Illumination



Photo credit: Jessica Andrews, flickr

Why is the back of cup illuminated?

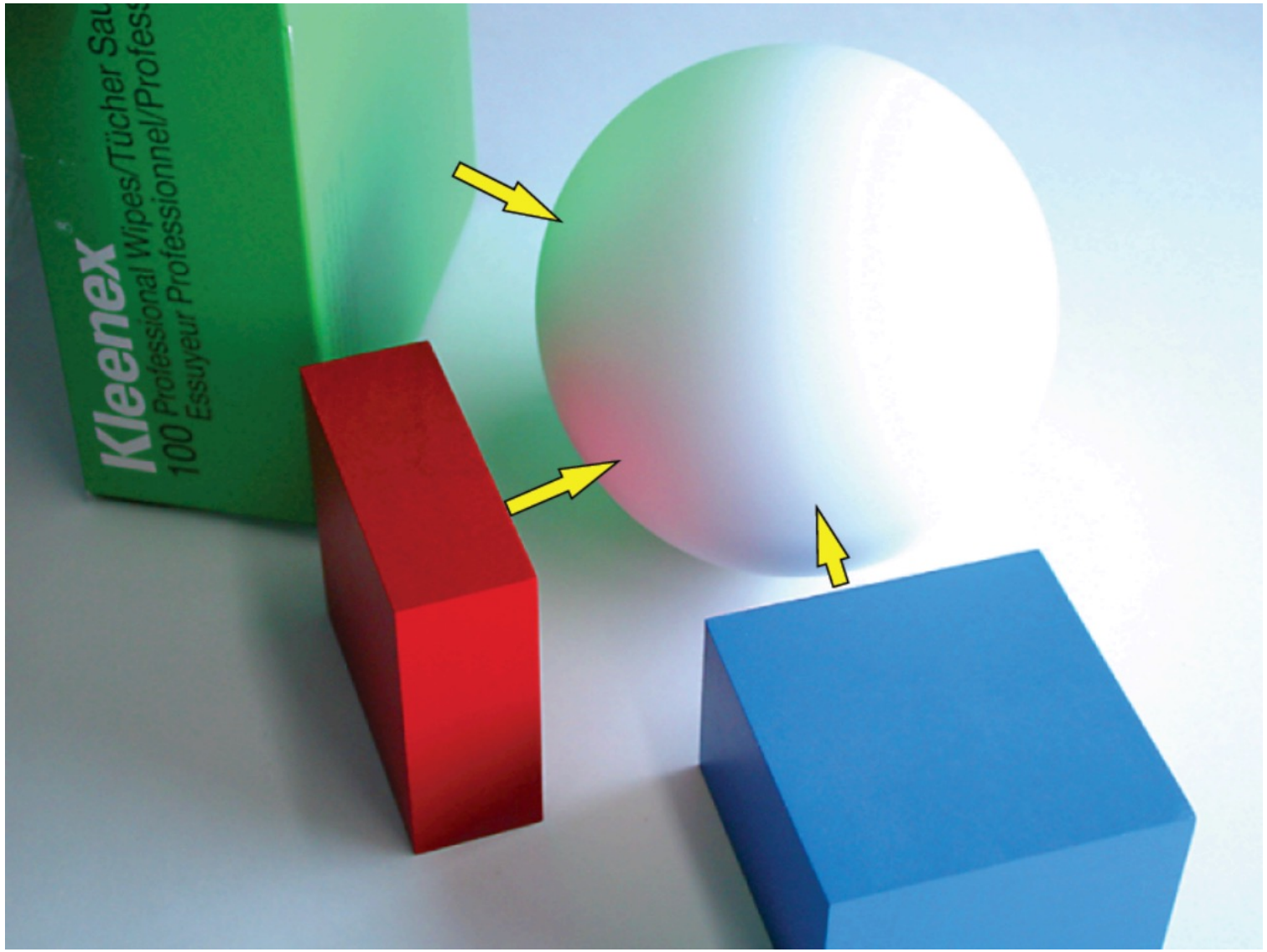
ambient illumination

$$f_a(\lambda) = k_a(\lambda) L_a(\lambda)$$

Phong Shading Model

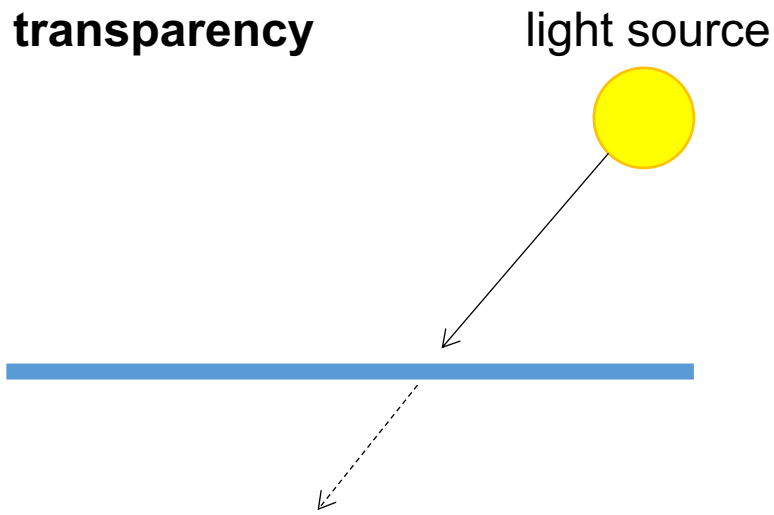
$$L_r(\hat{\mathbf{v}}_r; \lambda) = k_a(\lambda) L_a(\lambda) + k_d(\lambda) \sum_i L_i(\lambda) [\hat{\mathbf{v}}_i \cdot \hat{\mathbf{n}}]^+ + k_s(\lambda) \sum_i L_i(\lambda) (\hat{\mathbf{v}}_r \cdot \hat{\mathbf{s}}_i)^{k_e}$$

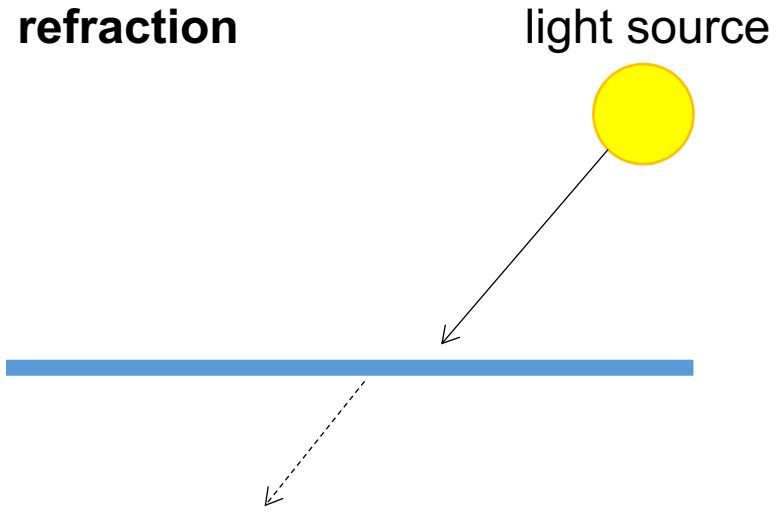
Interreflection



Credit to Koenderink and David Forsyth

Other Effects: Transparency and Refraction





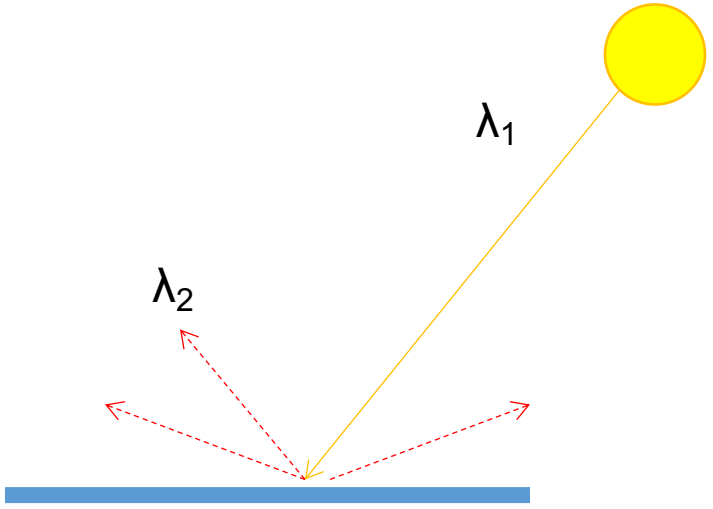


Other Effects: Fluorescence and Phosphorescence

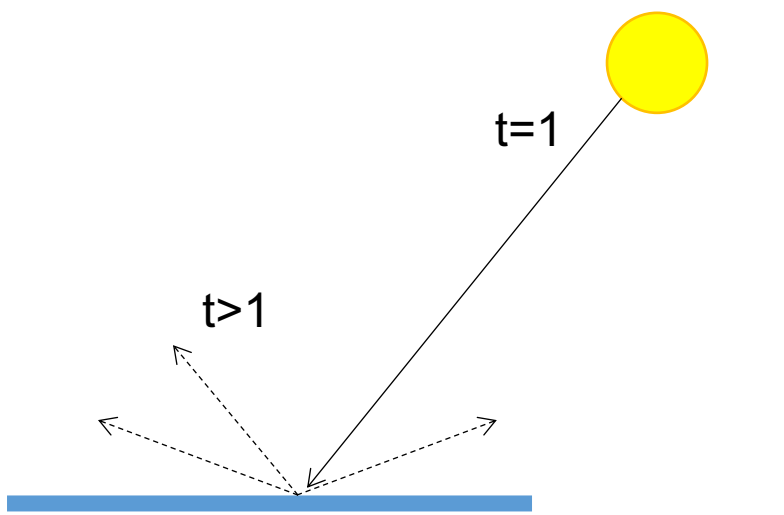




fluorescence



phosphorescence



Other Effects: Subsurface Scattering

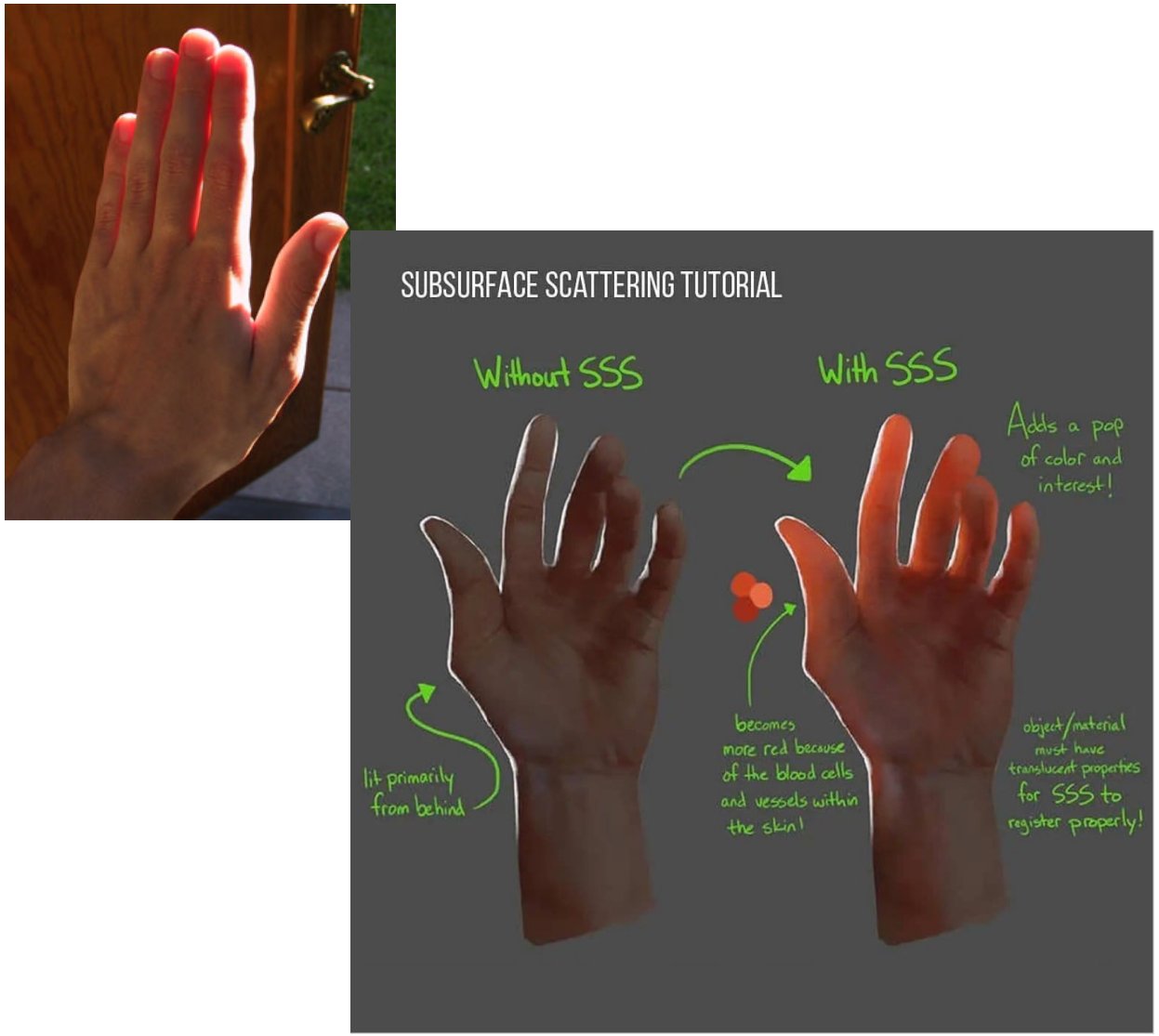
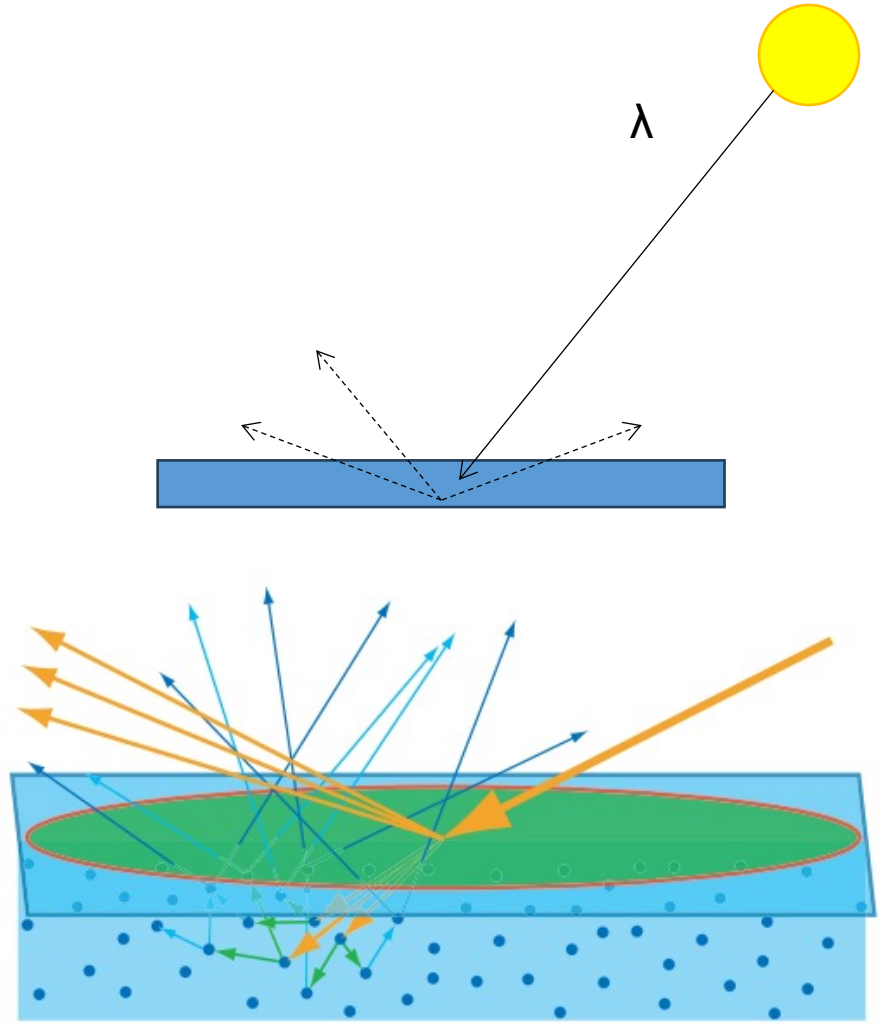


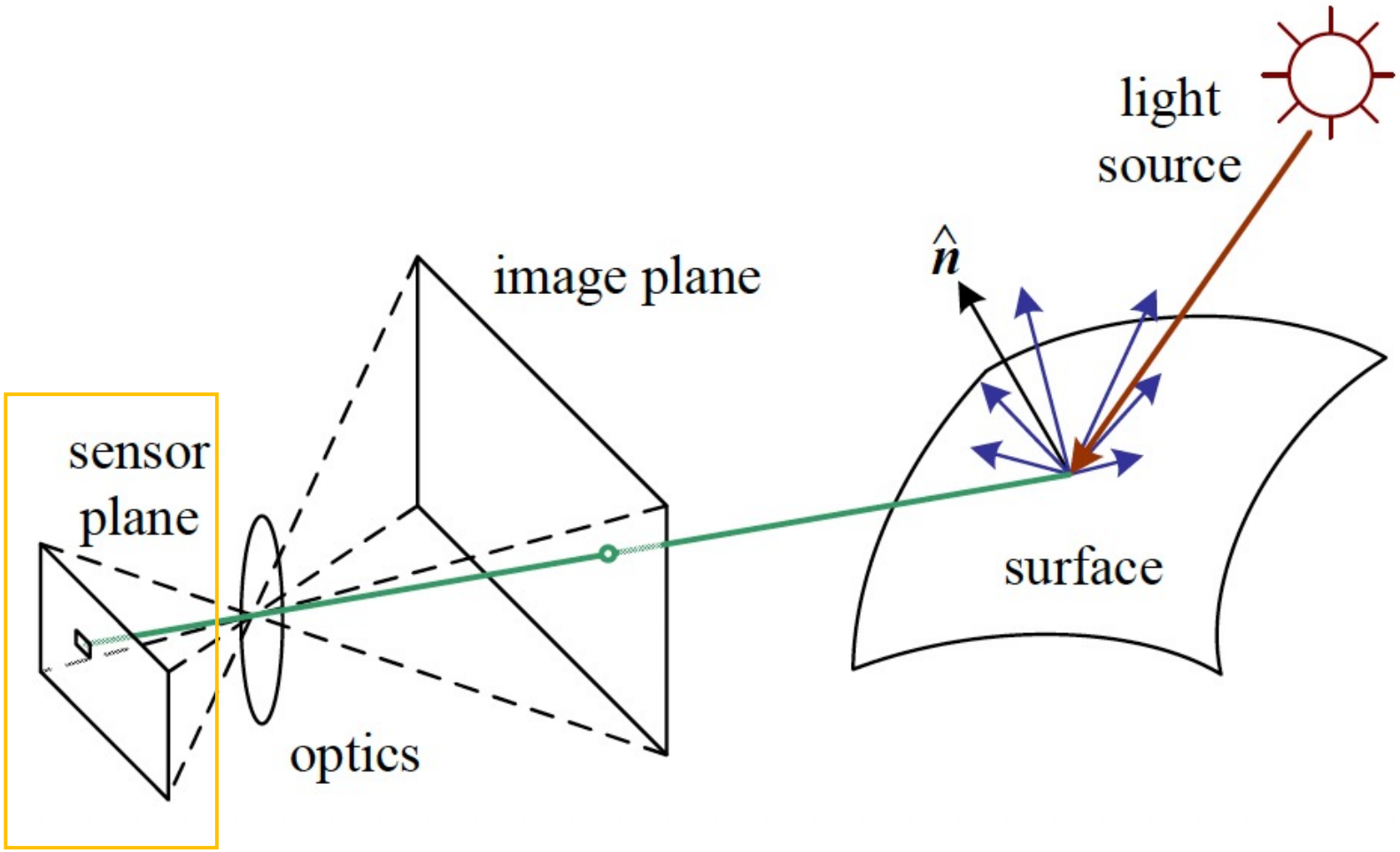
Figure from post

subsurface scattering (SSS) light source



https://therealmjp.github.io/posts/sss-intro/

How does the Sensor Process Received Intensity?



Let's finally check how the sensor processes received intensity

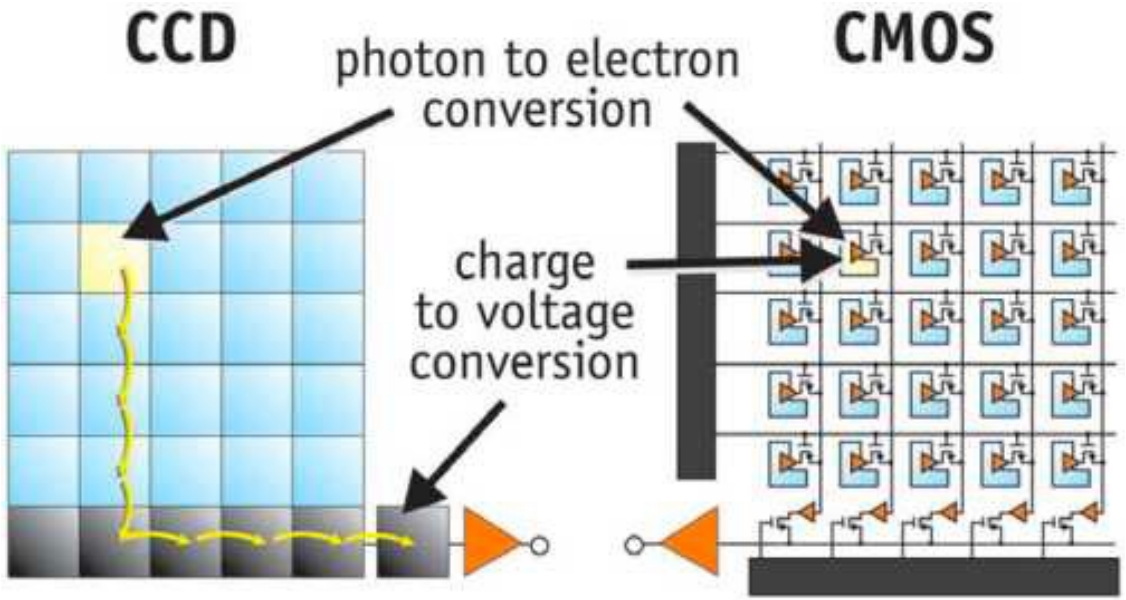
Szeliski, Figure 2.14

Digital Camera

How is a digital image formed?

Digital Camera





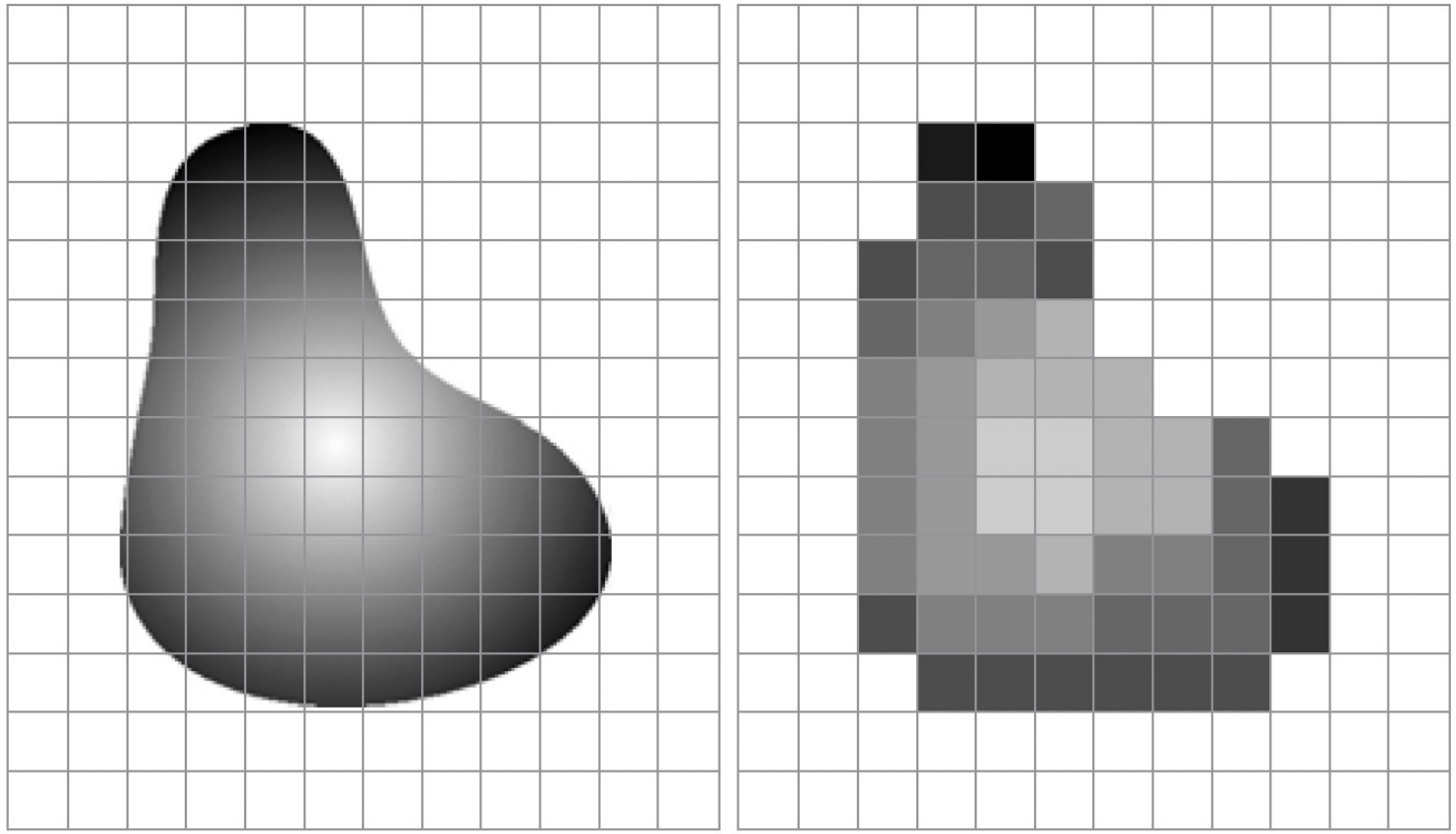
A digital camera replaces film with a sensor array

- Each cell in the array is light-sensitive diode that converts photons to electrons
- Two common types: Charge Coupled Device (CCD) and Complementary Metal-Oxide-Semiconductor (CMOS)

http://electronics.howstuffworks.com/digital-camera.htm

Credit to Steve Seitz

Image is a Pixel (**Pic**ture **El**ement) Array

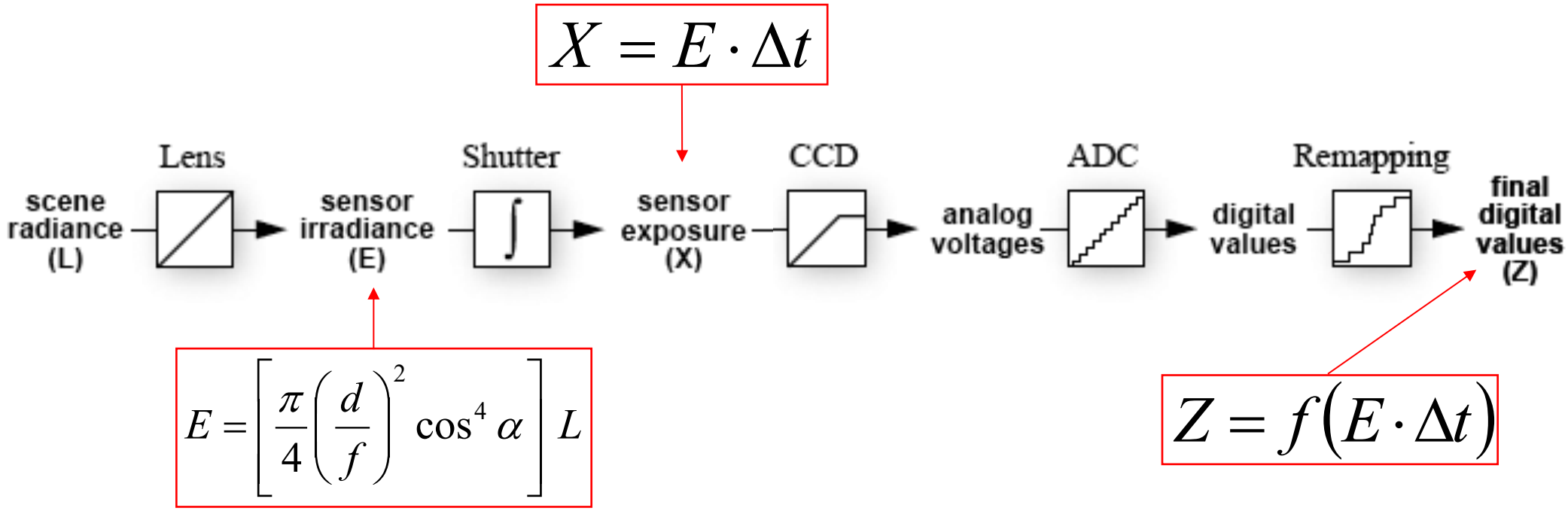


a b

FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

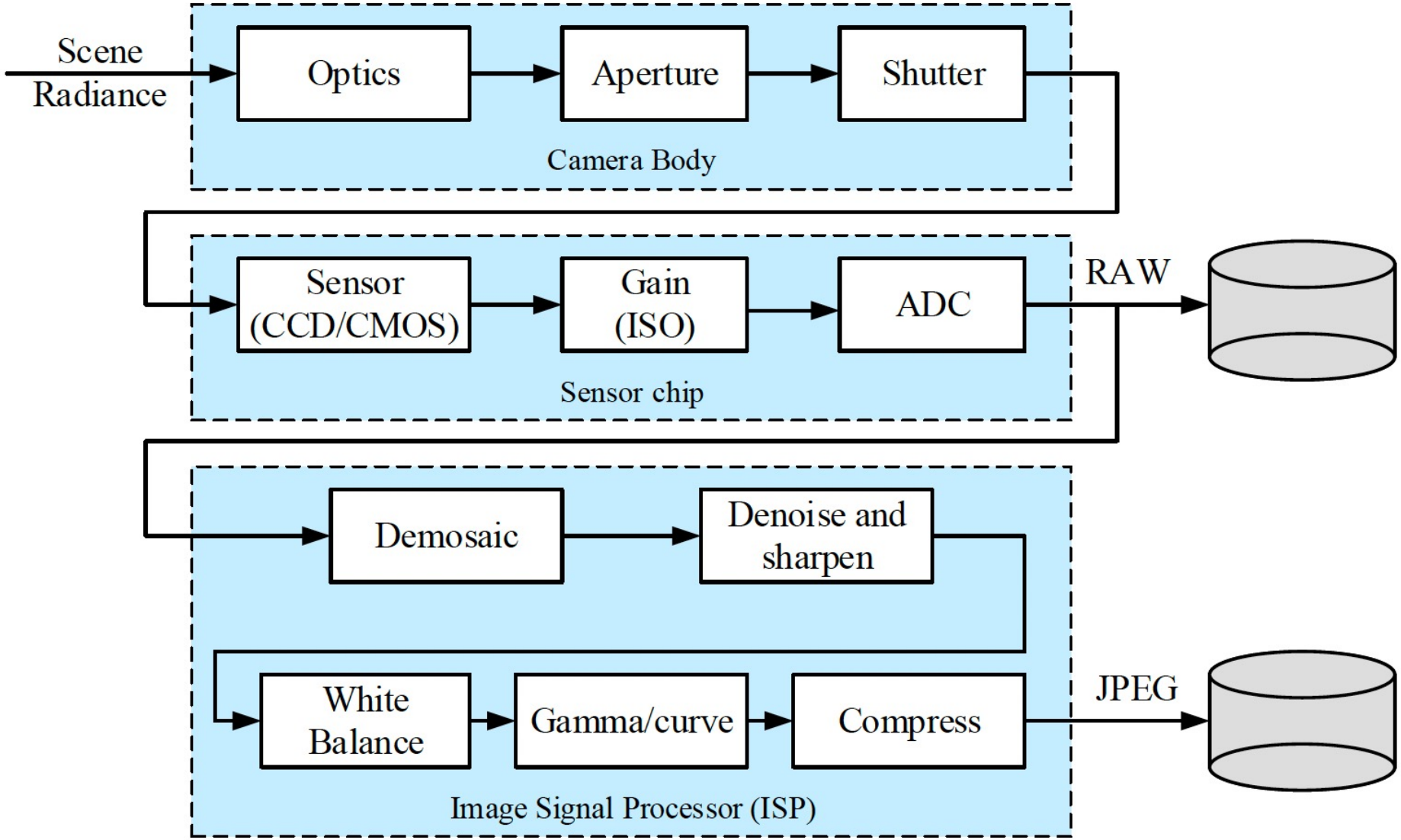
Figure from Digital Image Processing

From Light Rays to Pixel Values



- Camera response function: the mapping f from irradiance to pixel values
  - Enables us to create high dynamic range images
  - Useful if we want to estimate material properties
  - For more info: P. E. Debevec and J. Malik, *Recovering High Dynamic Range Radiance Maps from Photographs*, *SIGGRAPH 97*

Image Sensing Pipeline (ISP)



Szeliski, Ch 2.3, Figure 2.23

Photometric Stereo

Part II Inference from Shading

Shape from Shading



Luca della Robbia, *Cantoria*, 1438

Can we reconstruct the shape of an object based on shading cues?

Change the light, not the geometry

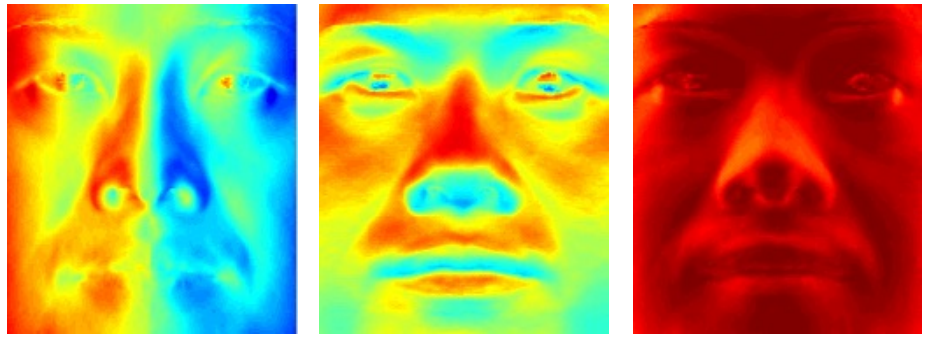
Input



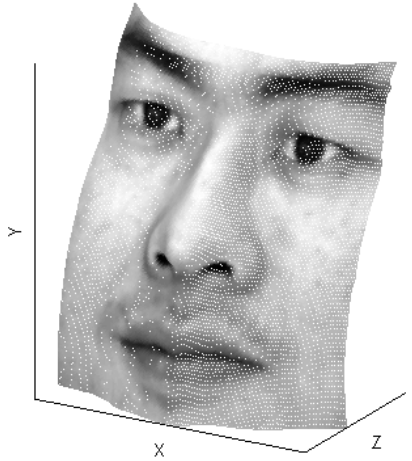
Recovered albedo



Recovered normal field



Recovered surface model



Photometric Stereo

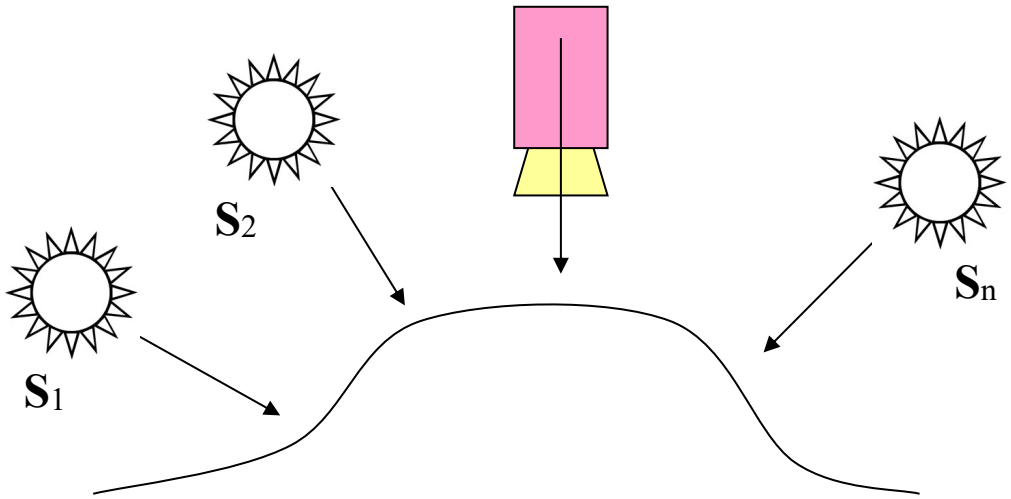
- Goal: reconstruct object shape and albedo

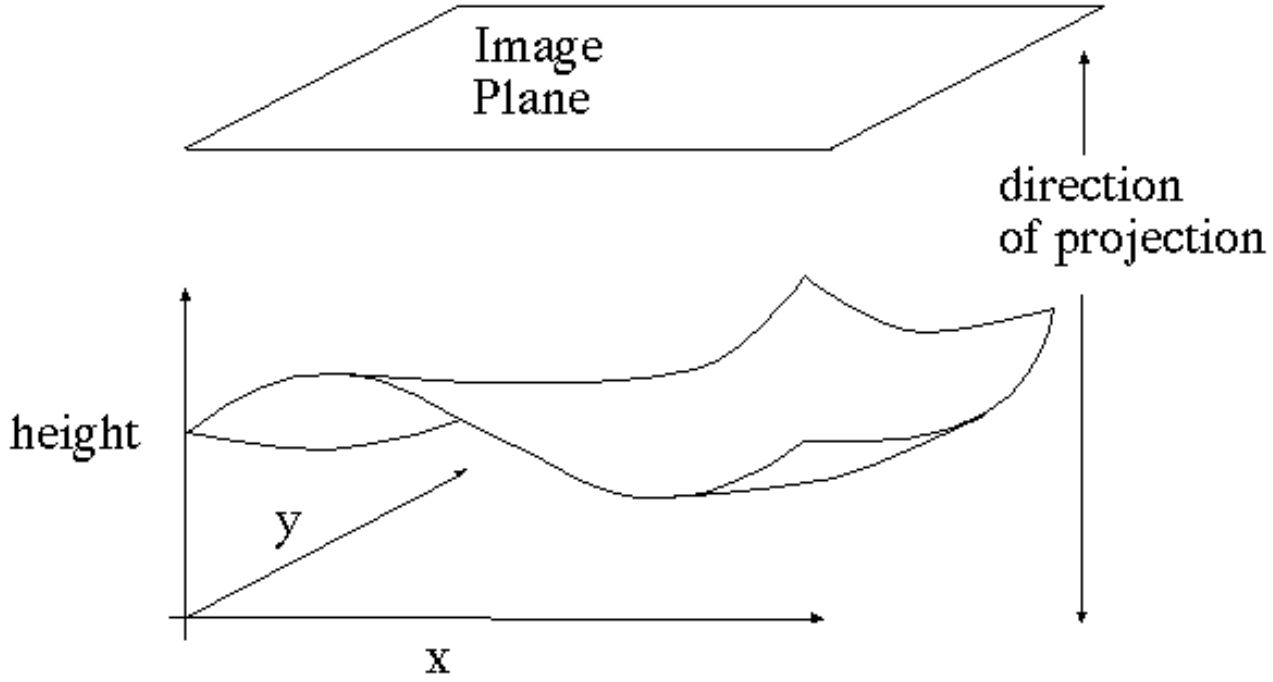
- Assumptions:
  - A *Lambertian* object
  - A *local shading model*
    - Each point on a surface receives light only from sources visible at that point
  - A set of *known* light source directions
  - A set of pictures of an object, obtained in exactly the same camera/object configuration but using different sources

F&P 2nd edition, sec. 2.2.4

Image Model

- **Known:** source vectors $\boldsymbol{S}_j$ and pixel values $\boldsymbol{I}_j(x, y)$
- **Unknown:** surface normal $\boldsymbol{N}(x, y)$ and albedo $\rho(x, y)$





F&P 2nd ed., sec. 2.2.4

Image Model

- **Known:** source vectors $\boldsymbol{S}_j$ and pixel values $\boldsymbol{I}_j(x, y)$
- **Unknown:** surface normal $\boldsymbol{N}(x, y)$ and albedo $\rho(x, y)$
- Assume that the response function of the camera is a linear scaling by a factor of k
- Lambert's law:

$$
\begin{aligned}
I_j(x,y) &= k\,\rho(x,y)\left(\mathbf{N}(x,y)\cdot\mathbf{S}_j\right) \\
&= \left(\rho(x,y)\mathbf{N}(x,y)\right)\cdot\left(k\mathbf{S}_j\right) \\
&= \mathbf{g}(x,y)\cdot\mathbf{V}_j
\end{aligned}
$$

F&P 2nd ed., sec. 2.2.4

Least Squares Problem

For each pixel, set up a linear system:

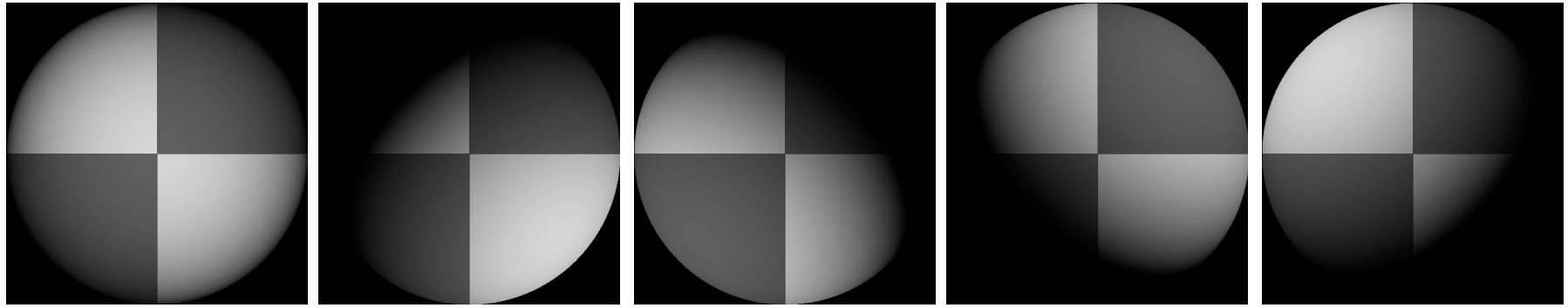
$$\begin{bmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{bmatrix} = \begin{bmatrix} \mathbf{V}_1^T \\ \mathbf{V}_2^T \\ \vdots \\ \mathbf{V}_n^T \end{bmatrix} \mathbf{g}(x,y)$$

(n × 1) known — (n × 3) known — (3 × 1) unknown

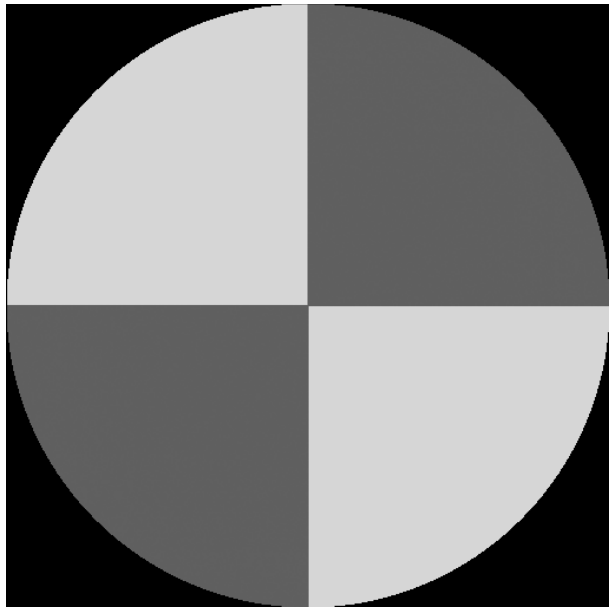
- Obtain least-squares solution for $\boldsymbol{g}(x,y)$, which we defined as $\rho(x,y)\boldsymbol{N}(x,y)$
- Since $\boldsymbol{N}(x,y)$ is the unit normal, $\rho(x,y)$ is given by the magnitude of $\boldsymbol{g}(x,y)$
- Finally, $\boldsymbol{N}(x,y) = \boldsymbol{g}(x,y)/\rho(x,y)$

F&P 2nd ed., sec. 2.2.4

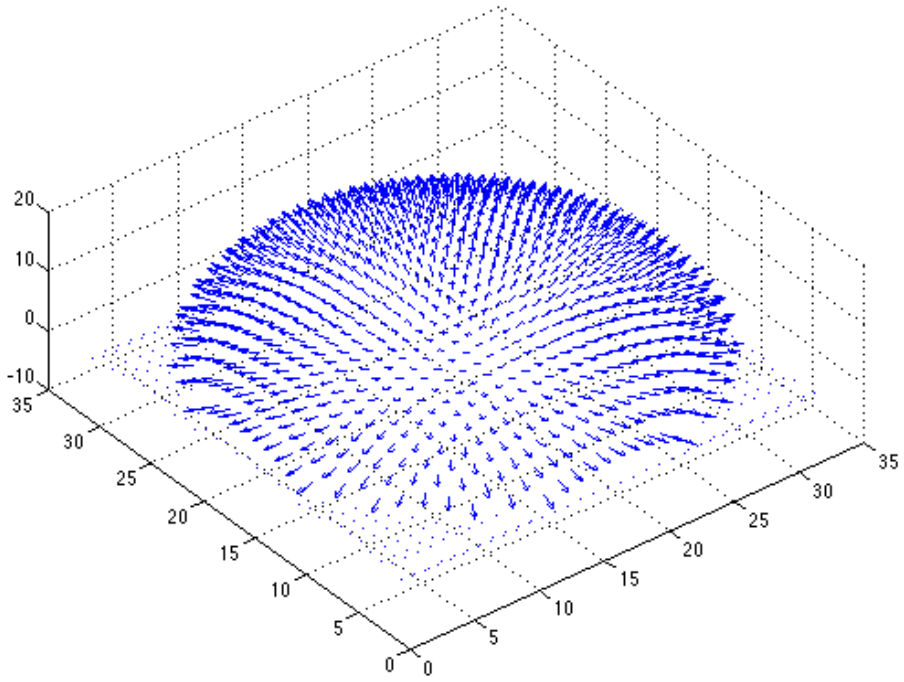
Synthetic Example



Recovered albedo



Recovered normal field



F&P 2nd ed., sec. 2.2.4

Recovering a Surface from Normals

- Recall: the surface is written as $P(x, y) = \big(x, y, z(x, y)\big)$
- The tangent plane is spanned by $t_x = \left(1, 0, \frac{\partial z}{\partial x}\right)$ and $t_y = \left(0, 1, \frac{\partial z}{\partial y}\right)$
- The normal $\boldsymbol{N} = (n_1, n_2, n_3)$ is a unit vector orthogonal to the tangent plane,
  so it is a normalized version of $t_x \times t_y = \left(-\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}, 1\right)$
- Thus, $\frac{\partial z}{\partial x} = -n_1/n_3, \frac{\partial z}{\partial y} = -n_2/n_3$

Recovering a Surface from Normals

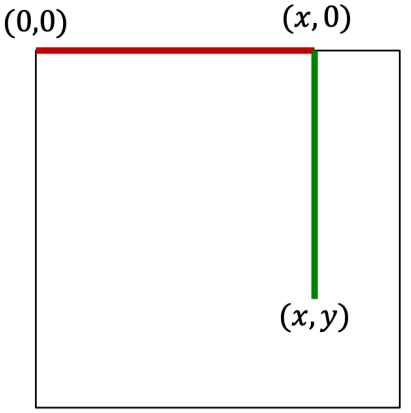
- Recall: the estimated vector g is written as

$$g(x,y) = \left(g_1(x,y), g_2(x,y), g_3(x,y)\right)$$

- Thus, we have $\frac{\partial z}{\partial x} = -\frac{g_1(x,y)}{g_3(x,y)}$, $\frac{\partial z}{\partial y} = -\frac{g_2(x,y)}{g_3(x,y)}$

- Then, we can recover the height z by integration

$$z(x,y) = \int_0^x \frac{\partial z}{\partial x}(u,0)du + \int_0^y \frac{\partial z}{\partial y}(x,v)dv + z(0,0)$$



Pseudo Codes

```
Input: Normal fields Nx, Ny, Nz
Output: Height map Z

# Normalize the normal vectors
Nx_normalized = Nx / Nz
Ny_normalized = Ny / Nz

# Initialize height map Z
Z = zeros_like(Nx)

# Perform numerical integration
for x in range(width):
    for y in range(height):
        if x > 0:
            Z[x, y] += Z[x-1, y] + Nx_normalized[x, y]
        if y > 0:
            Z[x, y] += Z[x, y-1] + Ny_normalized[x, y]
```

Limitations of Photometric Stereo

- **Camera**
  - linear response (RAW)
  - no saturation/noise
  - registered images (camera poses are calibrated)
- **Lighting**
  - calibrated light directions
  - no shadow
  - no interreflection
- **Material**
  - no specular reflection
  - Lambertian BRDF
- **Geometry**
  - no noise for simple integration

Summary

- Brightness depends jointly on light, shape, material, and camera
- BRDF describes reflectance
  - Lambertian reflection is a useful simplification
- Photometric stereo linearizes the inverse problem

Reading List

- **Forsyth & Ponce, *Computer Vision: A Modern Approach*, 2e**
  - Ch. 2 **Light and Shading**
    - 2.1 *Modelling Pixel Brightness*
    - 2.2.4 *Photometric Stereo: Shape from Multiple Shaded Images*
  - https://www.pearson.de/media/muster/toc/toc_9781292014081.pdf
- **Szeliski, *Computer Vision: Algorithms and Applications*, 2e**
  - Ch. 2 Image Formation
  - Focus: photometric image formation, lighting, reflectance, shading
- UC Berkeley CS180/280A — Capturing Light
- Woodham, Robert J.. “Photometric method for determining surface orientation from multiple images.” (1980).